

# CCT College Dublin

## Assessment Cover Page

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|-----------------------------|------------------------------------|
| <b>Module Title:</b>        | Client-Side Programming            |
| <b>Assessment Title:</b>    | CA1                                |
| <b>Lecturer Name:</b>       | Sam Weiss                          |
| <b>Student Full Name:</b>   | Bárbara Candido de Souza           |
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| <b>Date of Submission:</b>  | Sunday 31 <sup>st</sup> March 2024 |

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### Declaration

By submitting this assessment, I confirm that I have read the CCT policy on Academic Misconduct and understand the implications of submitting work that is not my own or does not appropriately reference material taken from a third party or other source. I declare it to be my own work and that all material from third parties has been appropriately referenced. I further confirm that this work has not previously been submitted for assessment by myself or someone else in CCT College Dublin or any other higher education institution.

# Report

Outlining the fundamentals of client-side programming for adding dynamic content this website.

## HTML

I've implemented a skeleton across all pages to ensure consistent structure and layout for organized content. This includes a header, navigation menu, the main content, sections dividing the other contents and the footer (MDN, 2023).

```
> skeleton.html > html > body > div > section
1  <html>
2  <head>
3    <meta>
4    <meta>
5    <title></title>
6    <link href="" rel="stylesheet" type="text/css">
7  </head>
8  <body>
9    <header>
10     <nav></nav>
11     <nav>
12       <ul>
13         <li><a href="">Home</a></li>
14         <li><a href="">About</a></li>
15         <li><a href="">Gallery</a></li>
16       </ul>
17     </nav>
18   </header>
19   <div>
20     <div></div>
21     <section>
22       <h1></h1>
23       <p></p>
24       <p></p>
25     </section>
26   </div>
```

Figure 1 Structure on HTML

```
27   <h2></h2>
28   <ul>
29     <li></li>
30     <li><strong></strong></li>
31     <li><strong></strong></li>
32     <li><strong></strong></li>
33     <li><strong></strong></li>
34   </ul>
35 </section>
36 <section>
37   <h2></h2>
38   <p></p>
39 </section>
40 <section>
41   <h3></h3>
42   <p></p>
43 </section>
44 </div>
45 <footer></footer>
46 <script src=""></script>
47 </body>
48 </html>
```

## CSS

In complement to the HTML structure, CSS styling has been applied to create an intuitive interface, facilitating user access and content consumption with a straightforward layout (W3School, 2024).

Adjustments such as font size and color, headings, spacing between content helps for a better visualization to the end user. The external CSS file applied to all pages ensures a consistency across all the website. (QuickRef.ME, 2024).

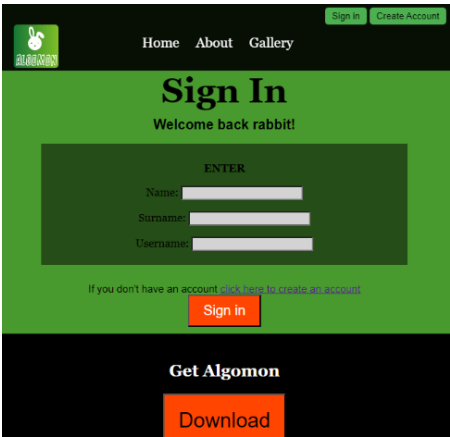


Figure 5 HTML with CSS

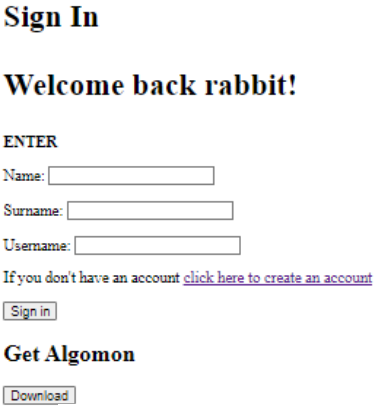
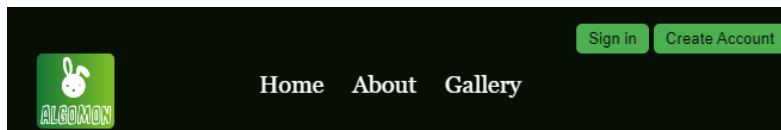


Figure 7 HTML without CSS

Examples of the CSS used in this website

## Site Navigation

Figure 9 Navigation Bar



For a more familiar feeling from the users, this website also tries to follow a certain standard. So that everyone knows how to use the website and don't feel lost while navigating. That also can be called the Jakob's Law of the Internet User Experience (Nielsen, 1999). Combining HTML, CSS and JavaScript the website gives the user a more comfortable feeling, making them stay longer to check more content.

The user can easily access all pages in the navigation bar. It contains 5 pages on the website: Home, About and Gallery located in the center and a Sign In and Create Account as buttons positioned in the right corner. Additionally, clicking the logo redirects users to the Home Page (dragosct, 2018).

```
header {
  position: sticky;
  top: 0;
  background-color: #000000e7;
  padding: 15px 0;
}

#login {
  text-align: right;
  color: black;
  padding: 0px 10px;
}

.loginButton a {
  text-decoration: none;
  color: inherit;
}

.loginButton {
  background-color: #4999A2E;
  color: black;
  padding: 5px 10px;
  border: 1px solid black;
  border-radius: 5px;
  cursor: pointer;
  display: inline;
  text-align: right;
}

header nav ul {
  list-style-type: none;
  text-align: center;
  margin: 0;
  padding: 15px;
}

header nav ul li {
  display: inline;
  margin-right: 20px;
}

header nav ul li a {
  color: #ffff;
  text-decoration: none;
  font-size: 20px;
}
```

Figure 10 Navigation bar CSS

## Fonts

Commonly used website fonts have been selected to enhance readability and convey clarity, contributing to visual consistency (Davidov, 2022).

```
body {
  margin: 0;
  font-family: Verdana, Arial, Helvetica, sans-serif;
  background-color: #4999A2E;
}
```

## Grid display for the gallery

A grid layout has been implemented for the gallery, offering an optimized visualization of images that dynamically adjusts based on the page size (Maheshwari, 2021).

```
.gallery {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
  grid-gap: 10px;
  margin-top: 20px;
  background-color: #00000080;
  margin-bottom: 20px;
}

.gallery img {
  width: 100%;
  height: auto;
}
```

Figure 12 Grid on CSS



Figure 15 Grid in the website

## Buttons

Flashy buttons have been incorporated to attract user attention and encourage interaction.

Figure 14 Button in the Website



Figure 13 Button on CSS

```
.buttondownload {
  background-color: #ff7f0e;
  padding: 20px 20px;
  font-size: 30;
}
```

## JavaScript - Event listeners

The JavaScript adds interactivity and more dynamic website enhancing the user experience. (SheCodes, 2023)

### Event Listener 1 – Mouseover and Mouseout

This is actually two Event Listeners, the **mouseover** and the **mouseout**. The user interacts with the navbar, on the three main pages and the other two pages placed as buttons in the corner.

```
//Links Change color
let listItems = document.getElementsByTagName("a");
let listItemsColours = [];

for (let item = 0; item < listItems.length; item++) {
  listItemsColours.push(listItems[item].style.color);

  listItems[item].addEventListener("mouseover", () => {
    listItems[item].style.color = "green";
  });

  listItems[item].addEventListener("mouseout", () => {
    listItems[item].style.color = listItemsColours[item];
  });
}
```

Figure 16 Event Listener 1 Code - Mouseover and Mouseout

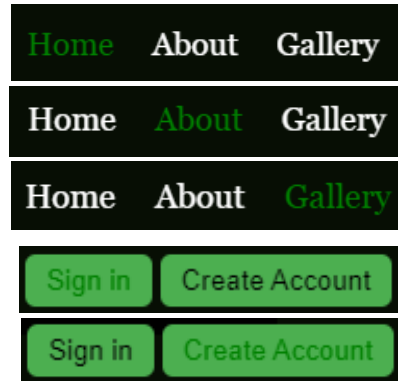


Figure 18 Event Listener 1 on website - Mouseover and Mouseout

### Event Listener 2 – Scroll

I wanted a different homepage display in the beginning where the navbar stays after the image and as you scroll down the image disappears and the navigation bar stays at the top. For that effect I could use an *Event Listener* on JavaScript to differentiate the main page from the other pages.

```
window.addEventListener('scroll', function() {
  scrollFunction();
});

function scrollFunction() {
  let scrollTop = document.documentElement.scrollTop;

  let navbar = document.getElementById("navbar");

  if (scrollTop > 80) {
    navbar.style.top = "0";
  } else {
    navbar.style.top = "-10px";
  }
}
```

Figure 20 Event Listener 2 code - Scroll

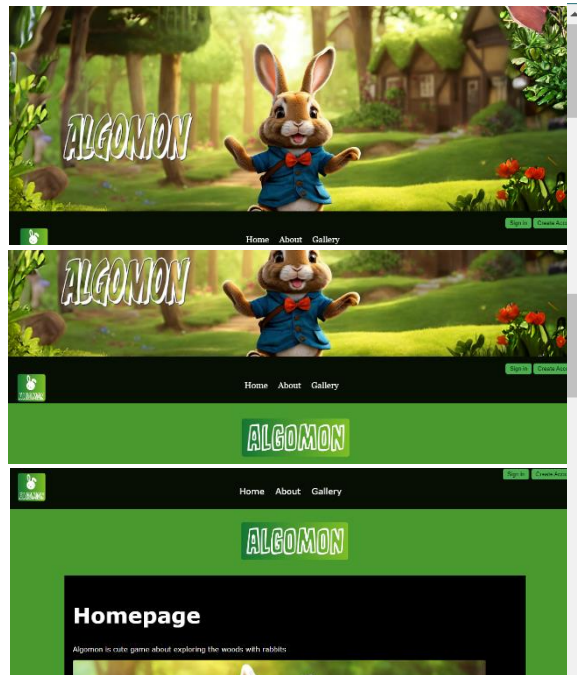


Figure 21 Event Listener 2 on website - Scroll

### Event Listener 3 – Click

A simple event listener that after the user click the button of Sign in or Download gets a message in JavaScript. (W3Schools, 2023)

Figure 23 Event Listener 3 on the website - Click

```
//Alert when clicking the button of signing in
document.getElementById("SignIn").addEventListener("click", function() {
  alert("Thank you for Signing In");
});

//Alert when clicking the button for downloading
document.getElementsByTagName("buttondownload").addEventListener("click", function() {
  alert("Downloading");
});
```

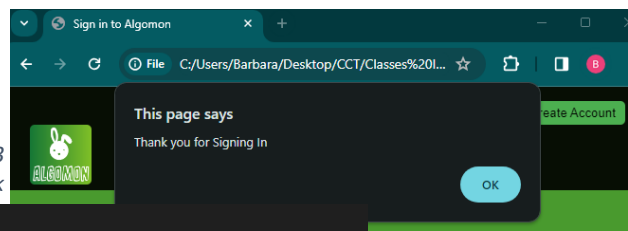


Figure 24 Event Listener 3 Code- Click

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